

Reducing Observational Astronomy Time Costs with Multi-Slit Spectrographs

Abdullah Korra, Dr. Katie Devine (The College of Idaho)

MWBest

1. Introduction

Our goal is to develop a data reduction pipeline for the KOSMOS (Kitt Peak Ohio State Multi-Object Spectrograph) instrument on the ARC 3.5m telescope (Fig. 1).



Figure 1: ARC 3.5m telescope at Apache Point Observatory.

2. Science Application

- **Objective:** Study extra-tidal stars from 20 inner Milky Way (Fig. 2) Globular Clusters (GCs) (Fig. 3).
- **GCs:** Densely packed groups of stars, gravitationally bound and moving together at a similar velocity.
- **Hypothesis:** Stripped GC stars significantly contribute to the inner Galaxy's metal-poor population.
- **Goal:** Better understand the Milky Way's formation history.

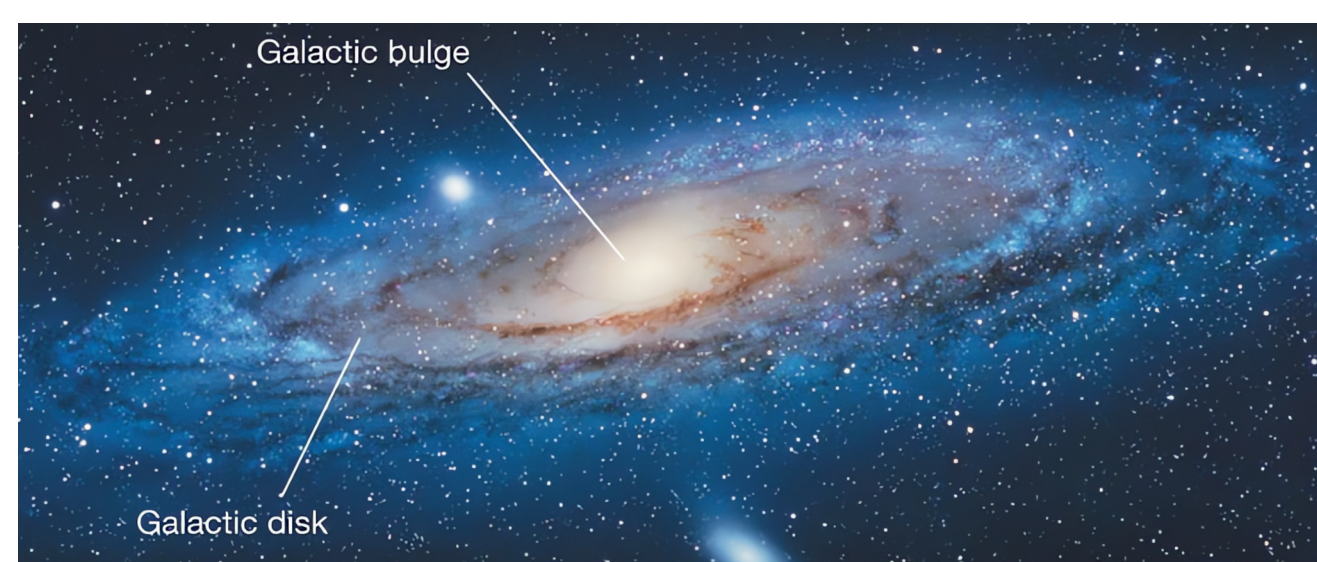


Figure 2: (Image Credit: ESA/Hubble & NASA, G. Piotto).

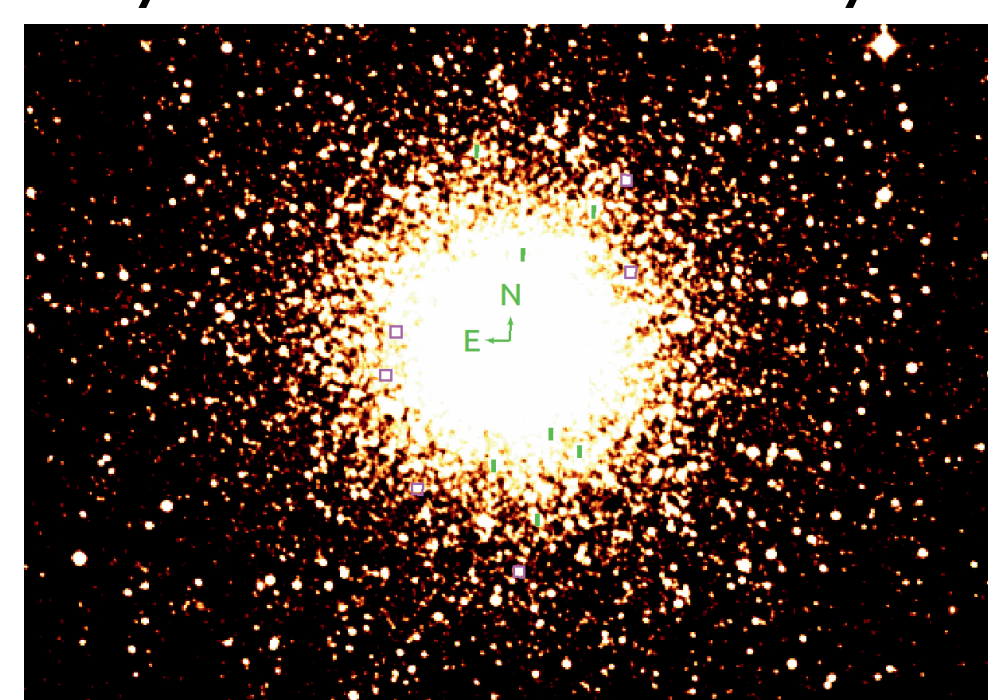


Figure 3: A globular cluster.

3. Previous Work

This project advanced our pipeline from *single-slit spectroscopy* (imaging one star at a time; fig. 4) to *multi-slit spectroscopy*, which images multiple stars simultaneously, resulting in a more efficient use of telescope time.

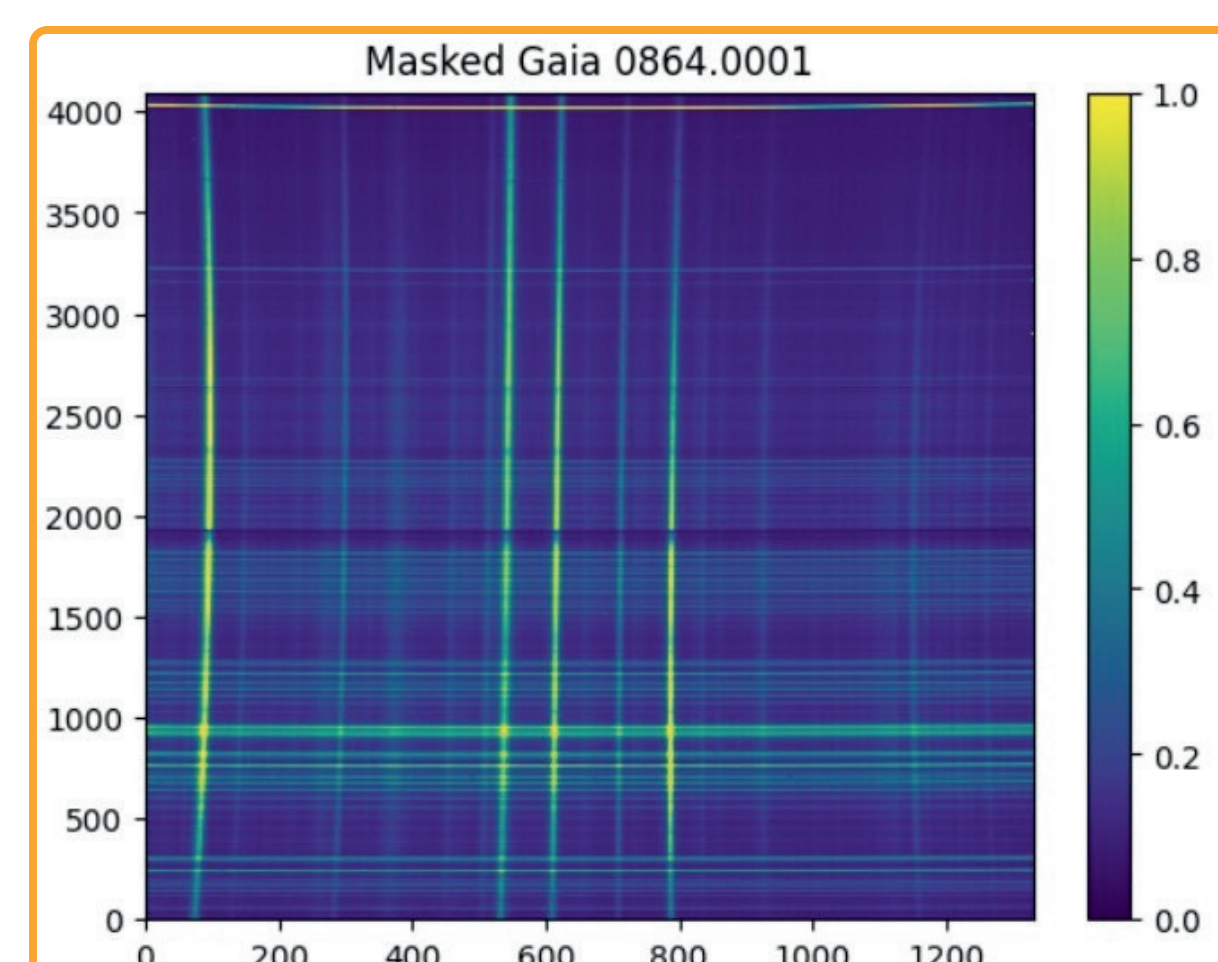


Figure 4: A single-slit spectrograph.

4. Methods

KOSMOS images show multiple stellar spectra per slit (Fig. 5), which the pipeline traces and extracts individually. Neon (Ne) spectra are then used to provide an accurate wavelength calibration for the stellar data (Figs. 6 & 8).

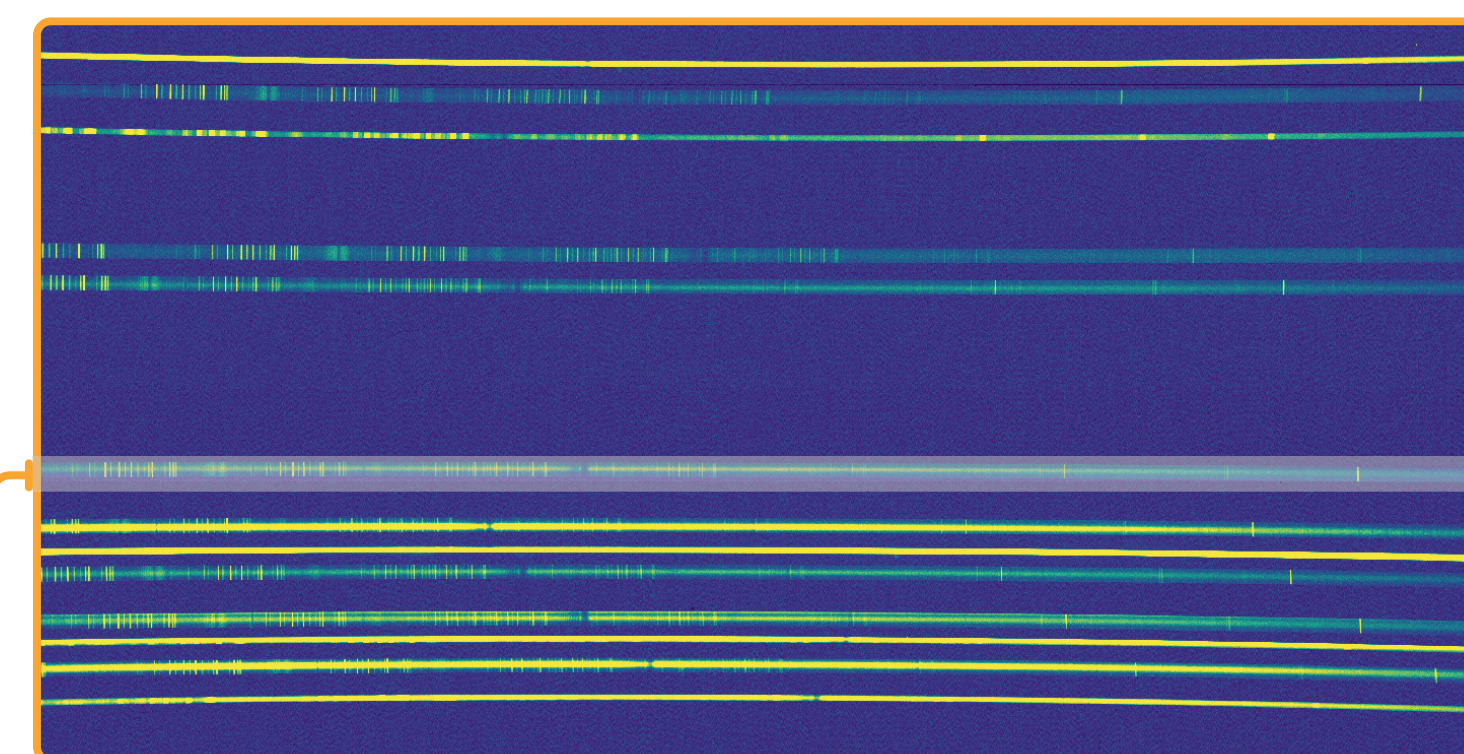


Figure 5: A calibrated reduced science image.

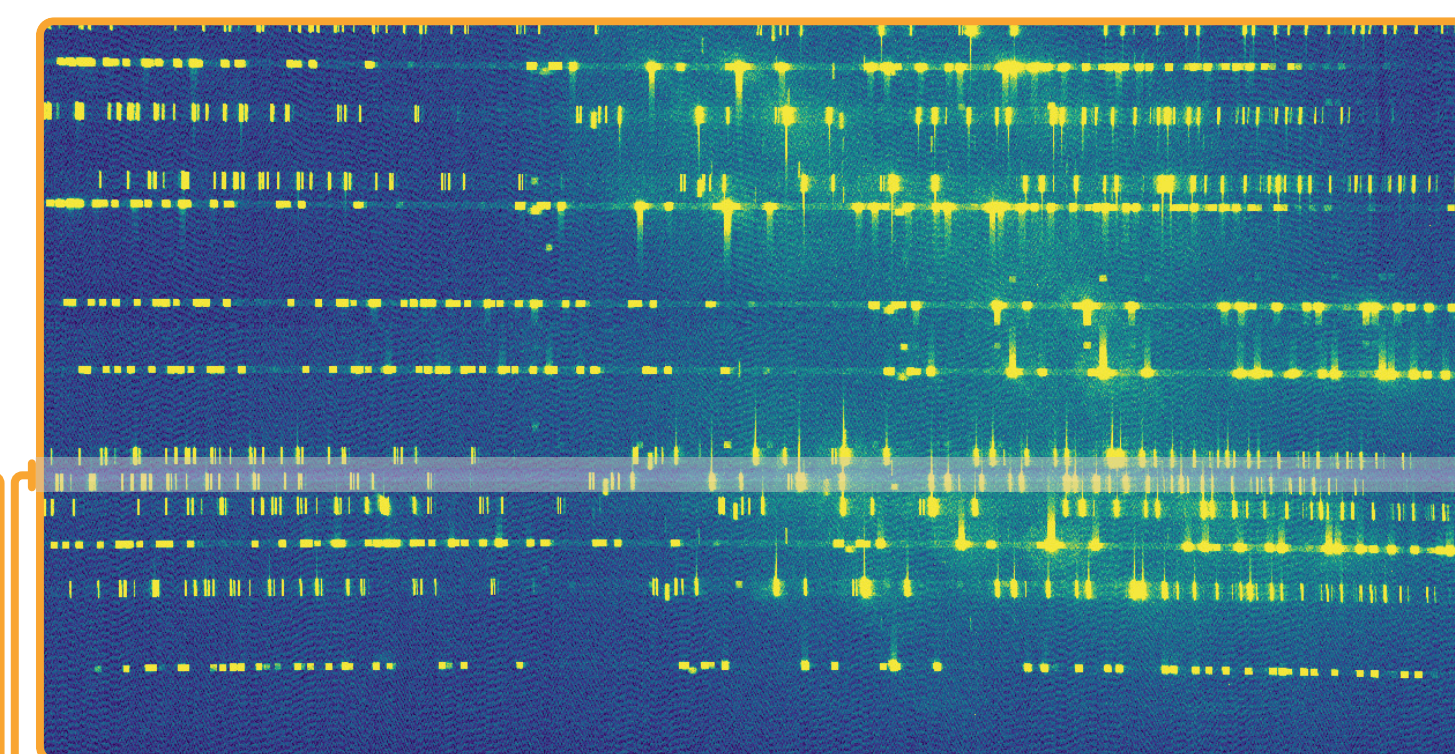


Figure 6: Neon lamp used for wavelength calibration.

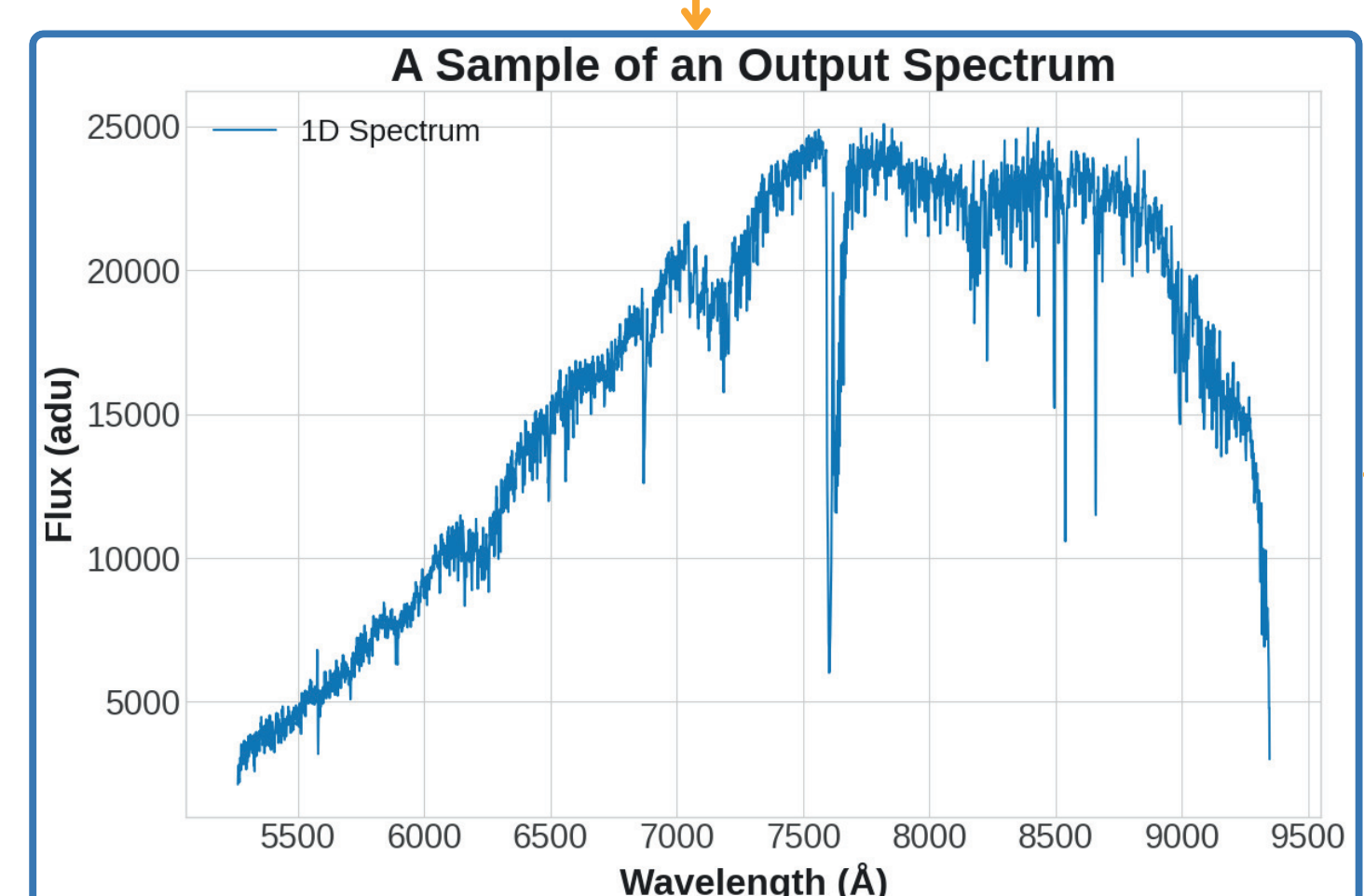
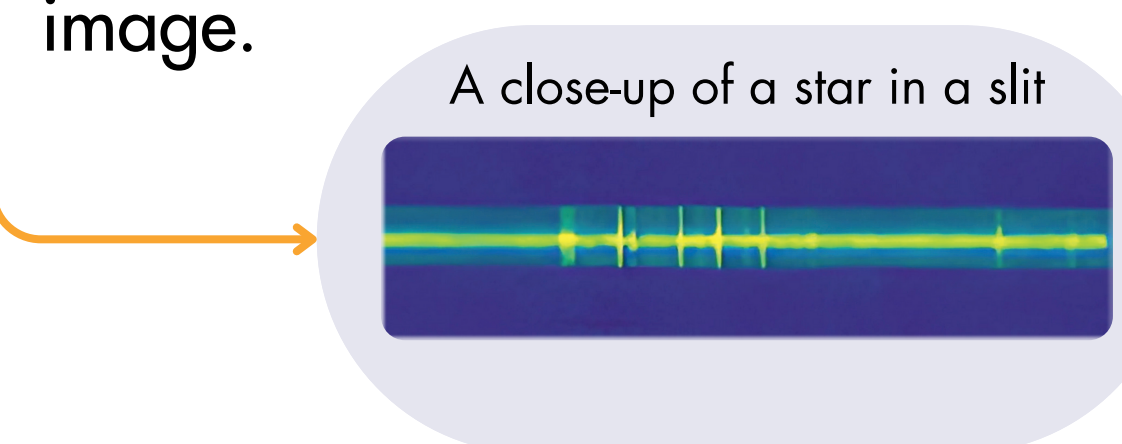


Figure 7: Extracted star spectrum.

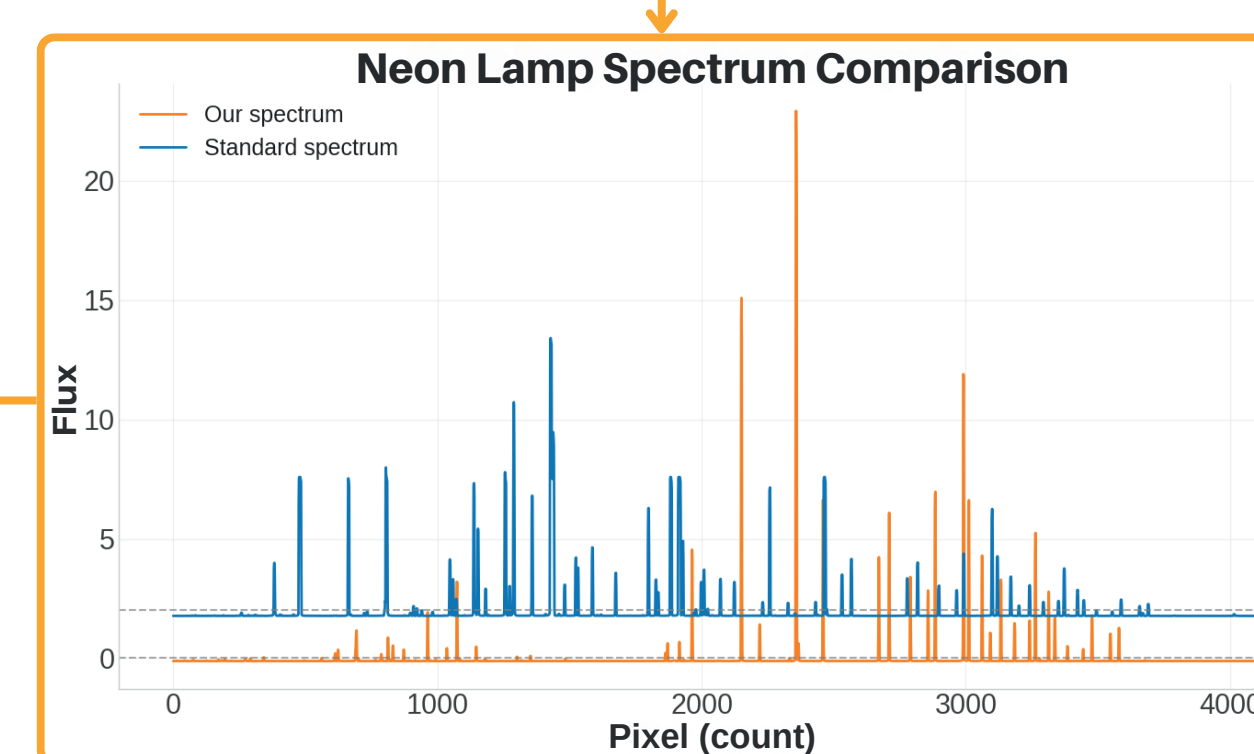


Figure 8: Wavelength solution obtained by matching the two spectra.

5. Results and Analysis

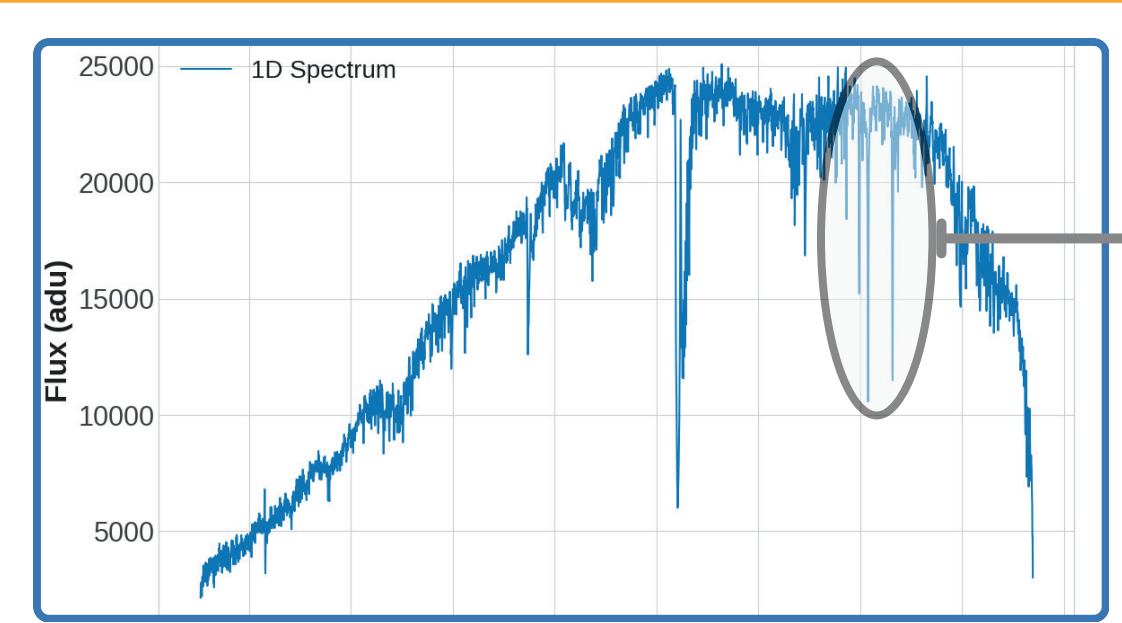


Figure 9: Extracted star spectrum.

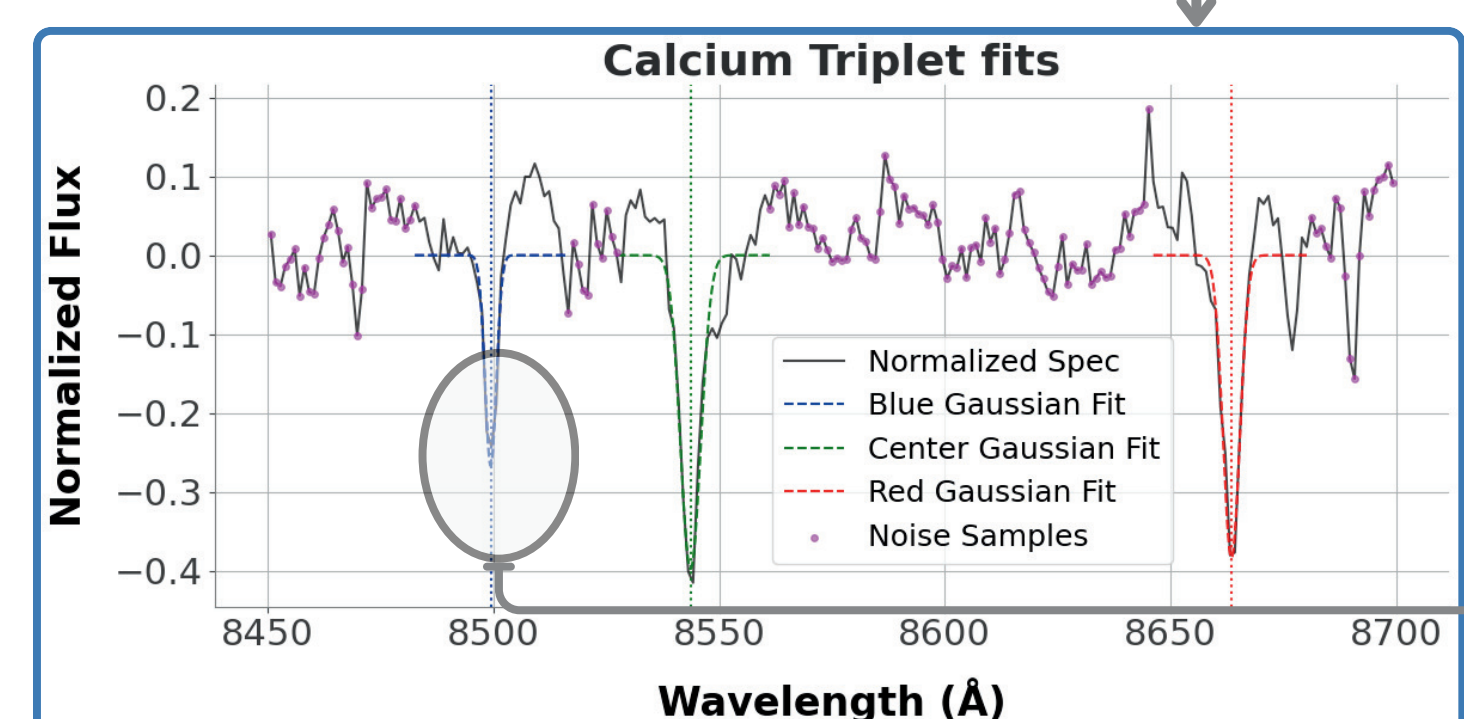


Figure 10: Gaussian fits of the calcium triplet lines.

Three fitting options are available (e.g., "Center Fit," "Original Trough"); selection is at the user's discretion.

- **Verification:** Method validated using stars with known radial velocities (RVs) from the APOGEE catalog.
- **RV Calculation:** RVs are derived from the Doppler shift of the Calcium Triplet lines (Fig. 10).
- **Comparison:** Our findings were compared to APOGEE catalog values (Fig. 12).

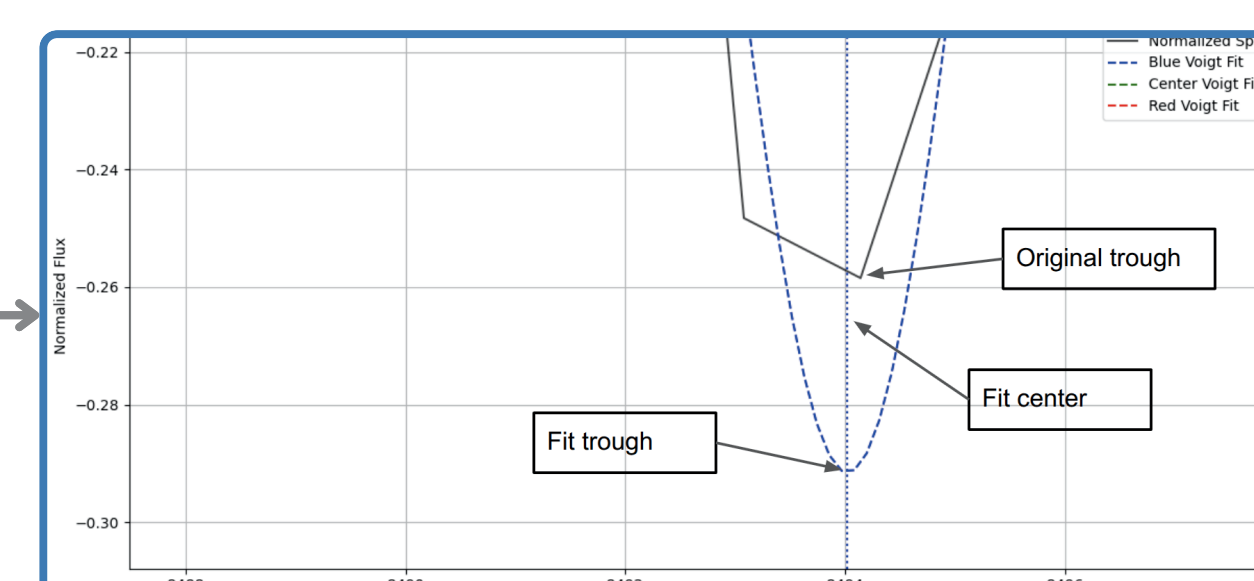


Figure 11: A closer look at a fit around a trough.

6. Results (RV Analysis)

Our RV measurements align well with APOGEE (Fig. 12), confirming the precision of our data reduction despite the instrument's low resolution. We also validated the method using cross-correlation RV calculations (Figs. 13, 14, & 15).

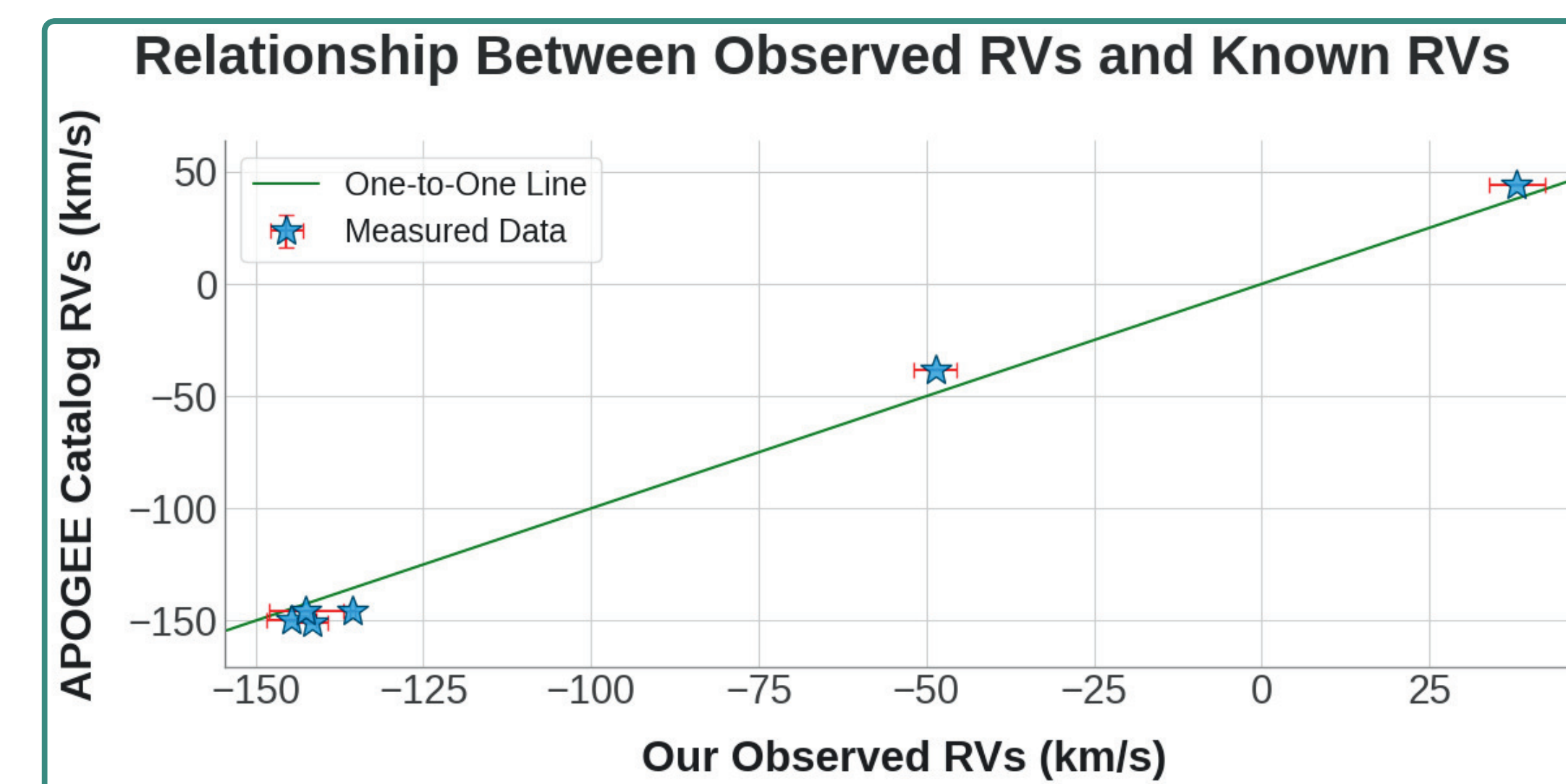


Figure 12: The plot shows our calculated RVs with a maximum difference of 10.6 km/s from the catalog values.

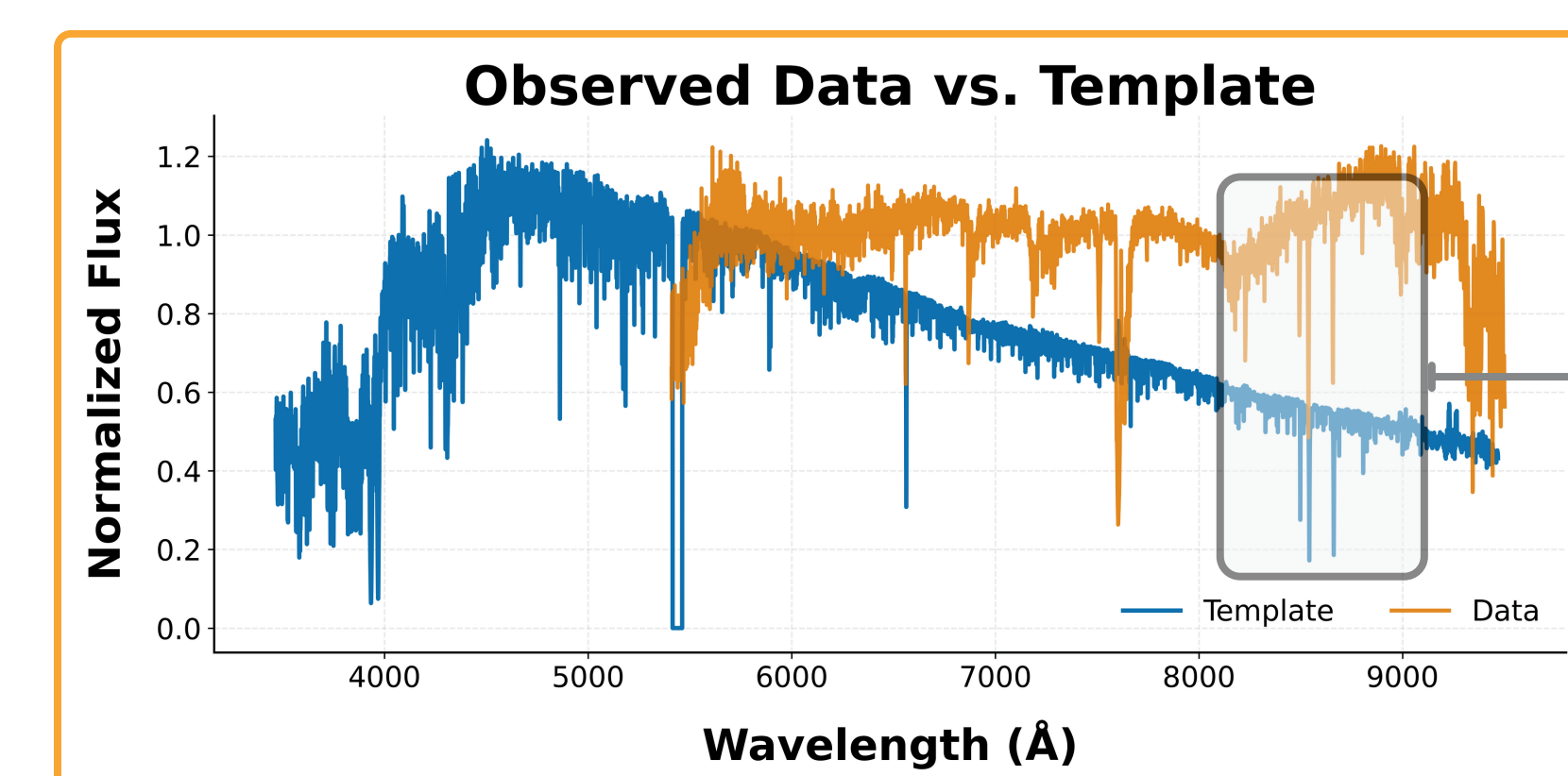


Figure 13: Visualization of spectra similarities.

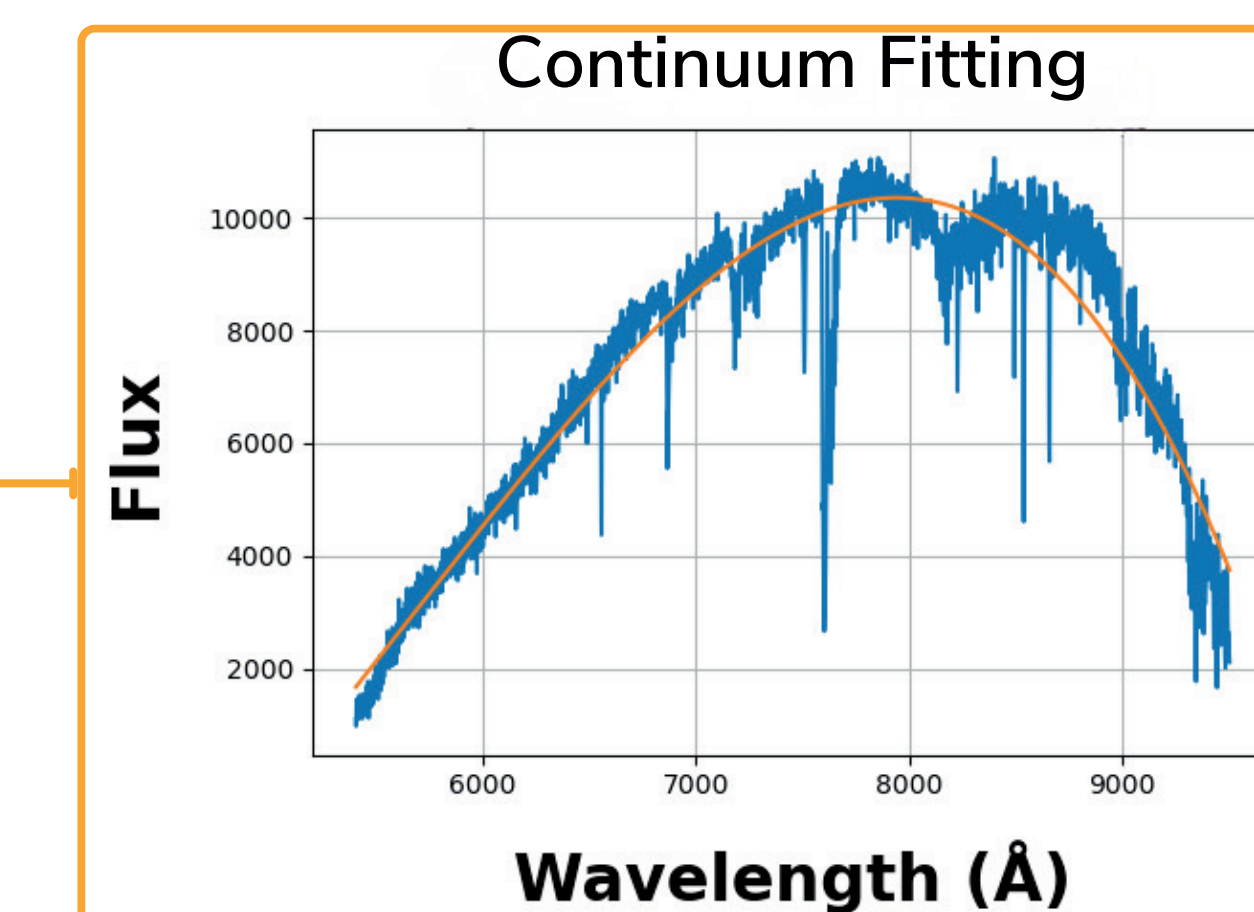


Figure 14: Normalization process.

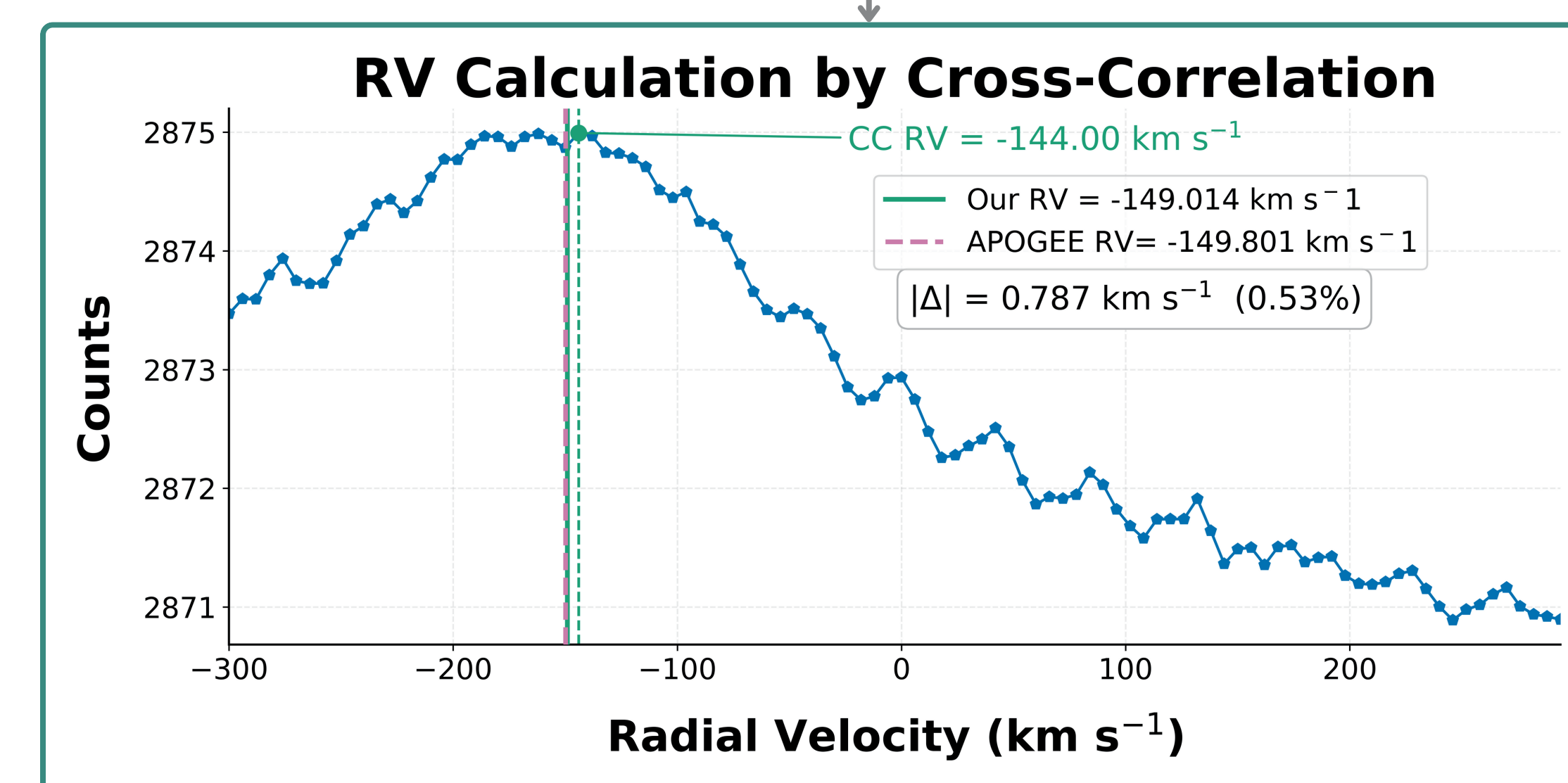


Figure 15: The curve shows spectral shifts within a selected region.

8. Acknowledgements

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